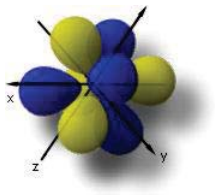
GENERAL CHEMISTRY

Chapter 1

Electronic Structure of Atoms



1

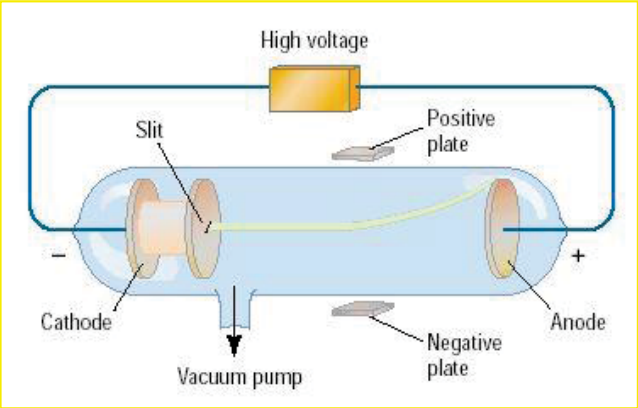
Defining the Atom

- ✓ **Elements are able to be subdivided** into smaller particles – these are the *atoms*, and they still have properties of that element.
- ✓ Dalton's atomic theory: **atoms are divisible** into subatomic particles: Electrons, protons, and neutrons.

2

Discovery of the Electron

In 1897, J.J. Thomson used a cathode ray tube to deduce the presence of a negatively charged particle: the **electron**

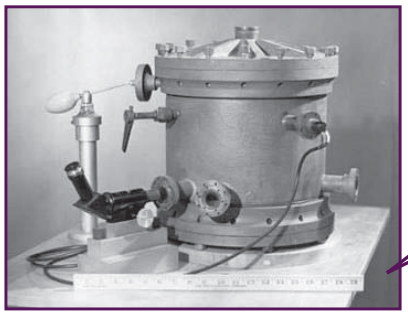


3

3

Mass of the Electron





Mass of the electron is 9.11×10^{-28} g

The oil drop apparatus

1916 – Robert Millikan determines the mass of the electron: 1/1840 the mass of a hydrogen atom; has one unit of negative charge .

4

4

Conclusions from the Study of the Electron

- ✓ All elements must contain identically charged electrons.
- ✓ Electrons have so little mass that atoms must contain other particles that account for most of the mass.
- ✓ Atoms are neutral, so there must be positive particles in the atom to balance the negative charge of the electrons.
- ✓ The "proton" - particles with a positive charge.
- ✓ The "neutron" – a particle with no charge.

5

5

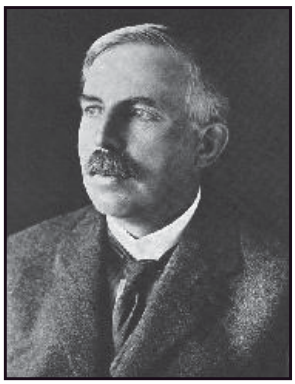
Conclusions from the Study of the Electron

Particle	Charge	Mass (g)	Location
Electron (e^-)	-1	9.11×10^{-28}	Electron cloud
Proton (p^+)	+1	1.67×10^{-24}	Nucleus
Neutron (n^o)	0	1.67×10^{-24}	Nucleus

6

6

Rutherford's Findings



- ✓ The nucleus is small
- ✓ The nucleus is dense
- ✓ The nucleus is positively charged

7

7

Complete Symbols

Contain the symbol of the element, the mass number and the atomic number.

Superscript → **Mass number**

Subscript → **Atomic number**

$$X$$

8

8

Symbols

- **Find each of these:**
  a) Number of protons
  b) Number of neutrons
  c) Number of electrons
  d) Atomic number
  e) Mass Number

$^{35}_{17}Cl$

$^{12}_{6}C$

$^{80}_{35}Br$

9

9

Symbols

- If an element has an atomic number of 34 and a mass number of 78, what is the:
  a) Number of protons
  b) Number of neutrons
  c) Number of electrons
  d) Complete symbol

$^{78}_{34}Se$

10

10

Symbols

- If an element has 91 protons and 140 neutrons what is the
  a) Atomic number
  b) Mass number
  c) number of electrons
  d) complete symbol

11

11

Symbols

- If an element has 78 electrons and 117 neutrons what is the
  a) Atomic number
  b) Mass number
  c) number of protons
  d) complete symbol

12

12

Isotopes

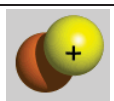
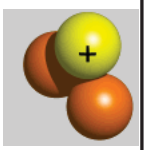


- Frederick Soddy (1877-1956) proposed the idea of isotopes in 1912.
- **Isotopes** are atoms of the **same element** having *different masses*, due to varying numbers of neutrons.

13

13

Naming Isotopes

Isotope	Protons	Electrons	Neutrons	Nucleus
Hydrogen–1 (protium)	1	1	0	
Hydrogen-2 (deuterium)	1	1	1	
Hydrogen-3 (tritium)	1	1	2	

14

14

EXAMPLE 5-1 ***Determination of Atomic Makeup***

Determine the number of protons, neutrons, and electrons in each of the following species. Are the members within each pair isotopes?

(a) $^{35}_{17}Cl$ and $^{37}_{17}Cl$ (b) $^{63}_{29}Cu$ and $^{65}_{29}Cu$

15

15

Atomic Mass

- We are more concerned with the average atomic mass.
- This is based on the abundance (percentage) of each variety of that element in nature.
  - We don't use grams for this mass because the numbers would be too small.
  - The mass of the isotope is expressed in atomic mass units (amu)

16

16

To calculate the average:

The atomic mass of element:

$$\overline{M} = \frac{M_1x_1 + M_2x_2 + M_3x_3 + \ldots + M_nx_n}{x_1 + x_2 + x_3 + \ldots + x_n}$$

EXAMPLE: **Oxygen ^{16}O (99.762 %), ^{17}O (0.038 %) and ^{18}O (0.200 %).**

→ ***Atomic mass*** *(in* amu)?

17

17

Atomic Masses

Atomic mass is the average of all the naturally occurring isotopes of that element.

Isotope	Symbol	Composition of the nucleus	% in nature
Carbon-12	^{12}C	6 protons 6 neutrons	98.89%
Carbon-13	^{13}C	6 protons 7 neutrons	1.11%
Carbon-14	^{14}C	6 protons 8 neutrons	<0.01%

Carbon = 12.011

18

18

Atomic Masses

EXAMPLE 5-2

Three isotopes of magnesium occur in nature. Their abundances and masses are listed in the following table. calculate the atomic weight of magnesium.

Isotope	% Abundance	Mass (amu)
$^{24}_{12}Mg$	78.99	23.98504
$^{25}_{12}Mg$	10.00	24.98584
$^{26}_{12}Mg$	11.01	25.98259

Atomic weight =24.30 amu

19

19

Modern Atomic Theory

20

20

The Uncertainty Principle

- **Heisenberg's Uncertainty Principle**: It is impossible to determine accurately both the momentum and the position of an electron (or any other very small particle) simultaneously.
- Δx is the uncertainty in position and Δmv is the uncertainty in momentum, then:

$$\Delta x \cdot \Delta m v \geq \frac{h}{4\pi}$$

21

21

Quantum Mechanics and Atomic Orbitals

- **Schrödinger** proposed an equation that contains both wave and particle terms.
- Solving the equation leads to wave functions.
- The **wave function** gives the **shape** of the electronic **orbital**.
- The square of the wave function (ψ^2) gives the probability of finding the electron. That is, gives the electron density for the atom.

22

22

The Schrödinger Wave Equation

- The Schrödinger wave equation in its time-dependent form for a particle of energy *E* moving in a potential *V* in one dimension is

$$i\hbar\frac{\partial\Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + V\Psi(x,t)$$

- The extension into three dimensions is

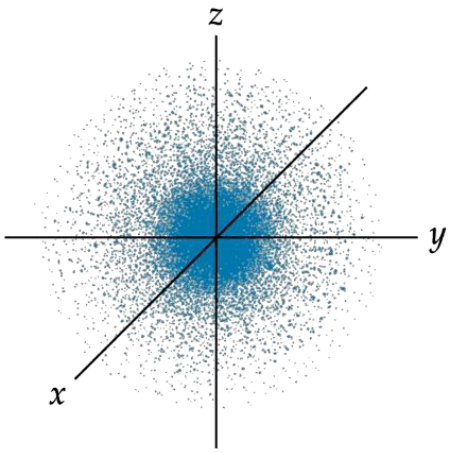
$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi}{\partial x^2} + \frac{\partial^2\Psi}{\partial y^2} + \frac{\partial^2\Psi}{\partial z^2}\right) + V\Psi(x,y,z,t)$$

where $i = \sqrt{-1}$ is an imaginary number.

23

23

Electron Density Distribution



Probability of finding an electron in a hydrogen atom in its ground state.

24

24

The Three Quantum Numbers

- **Schrödinger's equation requires 3 quantum numbers:**
- ✓ **Principal Quantum Number, *n (số lượng tử chính)*.**
- ✓ **Azimuthal Quantum Number, *l (số lượng tử orbital)*.**
- ✓ **Magnetic Quantum Number, m_l *(số lượng tử từ)***

25

25

The Principal Quantum Numbers

- **Principal Quantum Number, *n:***
- ✓ The same as Bohr's *n*.
- ✓ n, describes the main energy level, or shell, an electron occupies.
- ✓ As *n* becomes larger, the atom becomes larger
- ✓ *n* = 1, 2, 3…

26

26

The Azimuthal Quantum Number, *l*

- **Azimuthal Quantum Number or angular momentum quantum number, *l*:**
- ✓ Designates a sublevel, or specific shape of atomic orbital that an electron may occupy.
- ✓ May take integral values from 0 up to and including (n-1): *l*=0÷(*n* - 1).
- ✓ We give a letter notation to each value of *l*. Each letter corresponds to a different sublevel (subshell).

$$\ell = 0,\ 1,\ 2,\ 3,\ \ldots,\ (n-1)$$
$$s\quad p\quad d\quad f$$

27

27

The Azimuthal Quantum Number, *l*

Table 7.1 ▶ The Angular Momentum Quantum Numbers and Corresponding Letters Used to Designate Atomic Orbitals

Value of ℓ	0	1	2	3	4
Letter Used	*s*	*p*	*d*	*f*	*g*

$$E_{ns} < E_{np} < E_{nd} < E_{nf}$$

28

28

The Magnetic Quantum Number, m_l

- **Magnetic Quantum Number, m_l:**
  - ✓ Designates the specific orbital within a subshell.
  - ✓ Within each subshell, $\mathbf{m_l}$ may take any integral values from $-l$ through zero up to and including $+l$.
  - ✓ The maximum value of m_l depends on l.
  - ✓ $m_l = (-l) \div (+l)$.
  - ✓ Magnetic quantum numbers give the **3D orientation** of each orbital in space. ($m = -l \ldots 0 \ldots +l$)

29

29

The Magnetic Quantum Number, m_l

- ***For example:*** $l = 1$
  - ✓ *designates the p subshell,*
  - ✓ $\mathbf{m_l}$**:** -1, *0, and +1.*
  - ✓ *Thus, three distinct regions of space, called atomic orbitals, are associated with a p subshell.*
  - ✓ *We refer to these orbitals as the* p_x, p_y, *and* p_z *orbitals*

30

30

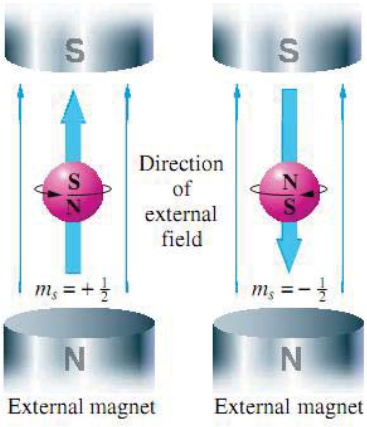
The Spin Quantum Number, m_s

- **Spin Quantum Number, m_s:**
- ✓ *Rrefers to the spin of an electron and the orientation of the magnetic field produced by this spin.*
- ✓ *For every set of n, l and m_l values, m_s can take the value + ½ or - ½.*

31

Electron Spin

- Two spots were found: **one with the electrons spinning in one direction and one with the electrons spinning in the opposite direction.**



- **m_s (spin quantum number) = ± ½.**

32

Orbitals and Quantum Numbers

TABLE 5-4 *Permissible Values of the Quantum Numbers Through* n = 4

n	ℓ	m_ℓ	m_s	Electron Capacity of Subshell = $4\ell + 2$	Electron Capacity of Shell = $2n^2$
1	0 (1*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	2
2	0 (2*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	8
	1 (2*p*)	$-1, 0, +1$	$\pm\frac{1}{2}$ for each value of m_ℓ	6	
3	0 (3*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	18
	1 (3*p*)	$-1, 0, +1$	$\pm\frac{1}{2}$ for each value of m_ℓ	6	
	2 (3*d*)	$-2, -1, 0, +1, +2$	$\pm\frac{1}{2}$ for each value of m_ℓ	10	
4	0 (4*s*)	0	$+\frac{1}{2}, -\frac{1}{2}$	2	32
	1 (4*p*)	$-1, 0, +1$	$\pm\frac{1}{2}$ for each value of m_ℓ	6	
	2 (4*d*)	$-2, -1, 0, +1, +2$	$\pm\frac{1}{2}$ for each value of m_ℓ	10	
	3 (4*f*)	$-3, -2, -1, 0, +1, +2, +3$	$\pm\frac{1}{2}$ for each value of m_ℓ	14	

33

Class Guided Practice Problem

- (a) For n = 4, what are the possible values of l? (b) For l = 2. What are the possible values of m_l? What are the representative orbital for the value of l in (a)?

Class Practice Problem

- (c) How many possible values for l and m_l are there when n = 3; n = 5?

34

Class Guided Practice Problem

EXAMPLE 5-9 *Electron Configurations and Quantum Numbers*

Write an acceptable set of four quantum numbers for each electron in a chlorine atom.

Electron	n	ℓ	m_ℓ	m_s	e^- Configuration
1, 2	1	0	0	$\pm\frac{1}{2}$	$1s^2$
3, 4	2	0	0	$\pm\frac{1}{2}$	$2s^2$
5–10	2	1	−1	$\pm\frac{1}{2}$	$2p^6$
	2	1	0	$\pm\frac{1}{2}$	
	2	1	+1	$\pm\frac{1}{2}$	
11, 12	3	0	0	$\pm\frac{1}{2}$	$3s^2$
13–17	3	1	−1	$\pm\frac{1}{2}$	$3p^5$
	3	1	0	$\pm\frac{1}{2}$	
	3	1	+1	$+\frac{1}{2}$ or $-\frac{1}{2}$*	

**The 3p orbital with only a single electron can be any one of the set, not necessarily the one with $m_\ell = +1$.*

35

Class Guided Practice Problem

EXAMPLE 5-11 *Unpaired Electrons*

Determine the number of unpaired electrons in an atom of tellurium, Te.

(Z=52)

Solution

The notation s^2p^4 is a short representation for s ⇅ p ⇅ ↑ ↑. This shows that an atom of Te contains two unpaired electrons.

36

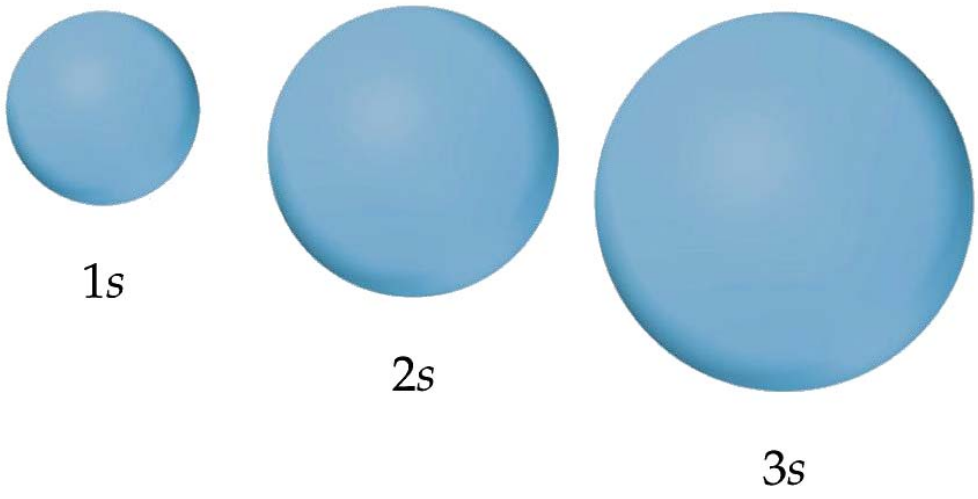
Representations of Orbitals
The s-Orbitals

- All s-orbitals are spherical.
- As n increases, the s-orbitals get larger.

37

37

The s-Orbitals

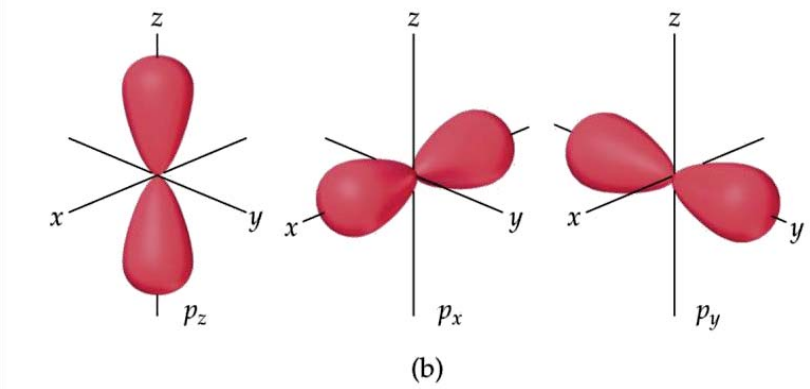


38

38

The p-Orbitals

- There are three p-orbitals p_x, p_y, and p_z.
- The three p-orbitals lie along the x-, y- and z- axes of a Cartesian coordinates.



39

39

The p-Orbitals

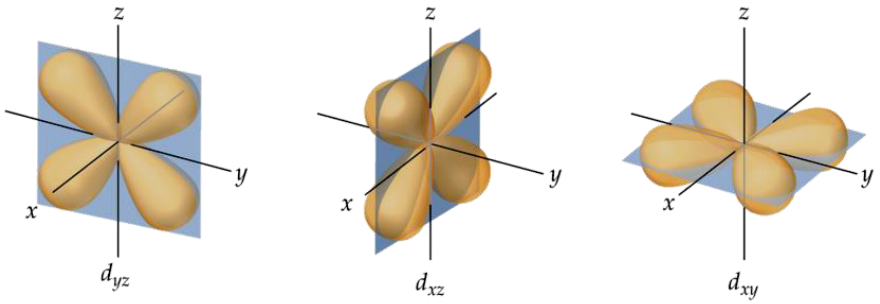
- The letters correspond to allowed values of m_l of -1, 0, and +1.
- The orbitals are dumbbell shaped.
- As n increases, the p-orbitals get larger.
- All p-orbitals have a node at the nucleus.

40

40

The d and f-Orbitals

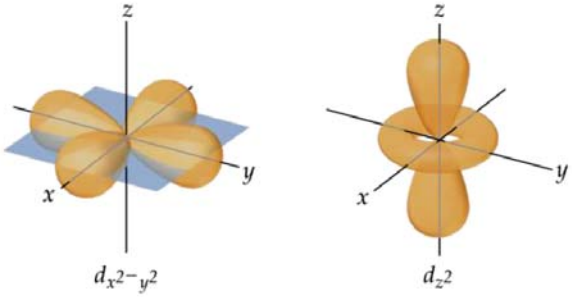
- **There are five d and seven f-orbitals.**
- Three of the d-orbitals lie in a plane bisecting the x-, y- and z-axes.



41

The d-Orbitals

- Two of the d-orbitals lie in a plane aligned along the x-, y- and z-axes.



- Four of the d-orbitals (d_{xy}, d_{xz}, d_{yz}, $d_{x^2-y^2}$) have four lobes each.
- Other d-orbital (d_{z^2}) has two lobes and a collar.

42

Electron Configurations

- **Electron configurations** tells us in which orbitals the electrons for an element are located.
- Three rules:
  - Aufbau principle (*German aufbauen, "to build up")*
  - **Pauli's Exclusions Principle**
  - **Hund's rule**.

43

43

Aufbau principle

- Electrons fill orbitals starting with lowest *n* and moving upwards.

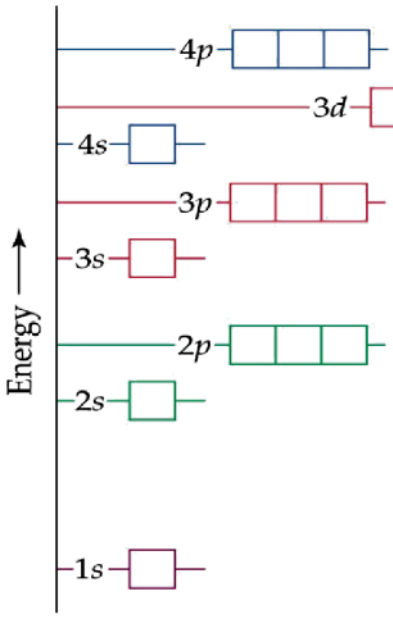
Two general rules help us to predict electron configurations:

- Electrons are assigned to orbitals in order of increasing value of (n+l).
- For subshells with the same value of (n+l), electrons are assigned first to the subshell with lower n.

44

44

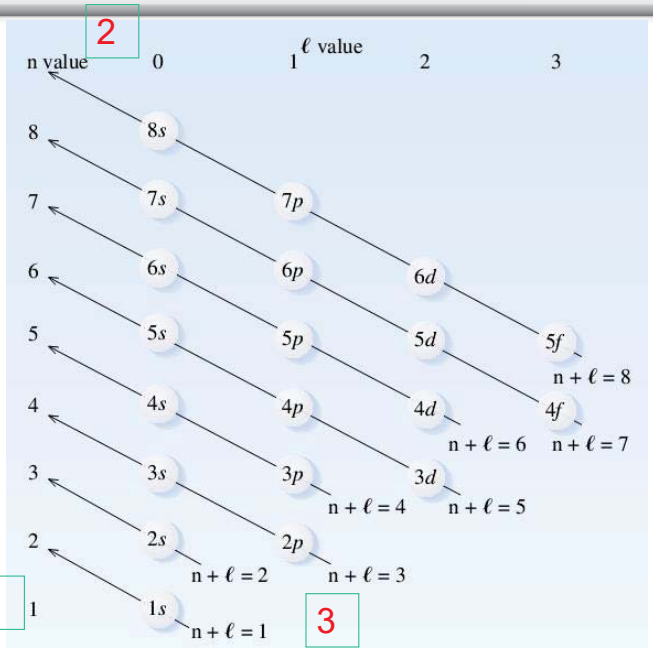
Aufbau diagram



1. The largest energy gap is between the 1s and 2s orbital
2. The gap between np and (n+1)s is fairly large.
3. The gap between (n-1)d and ns is quite small.
4. The gap between (n-2)f and ns is even smaller.

45

45



When subshells have the same (n+l), the subshell with the lower n fill first.

46

46

The Pauli Exclusion Principle

- **Pauli's Exclusions Principle**: no two electrons can have the same set of 4 quantum numbers.
  - Therefore, two electrons in the same orbital must have opposite spins.

Example: (He, Z=2)

Quantum numbers	n	ℓ	m_ℓ	s
e_1	1	0	0	1/2
e_2	1	0	0	-1/2

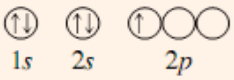


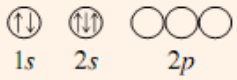
47

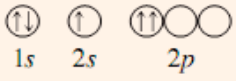
47

The Pauli Exclusion Principle

Example:
Which of the following orbital diagrams or electron configurations are possible and which are impossible, according to the Pauli exclusion principle? Explain.

a. 

b. 

c. 

d. $1s^3 2s^1$

e. $1s^2 2s^1 2p^7$

f. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$

48

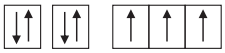
48

Hund's Rule

❖**Hund's rule:** The lowest energy arrangement of electron in a subshell is obtained by putting electrons into separate orbitals of the subshell with the same spin before pairing electrons.

- Electrons fill each orbital singly before any orbital gets a second electron.

Example: *The Orbital diagram for the ground state of the Nitrogen (Z=7)*

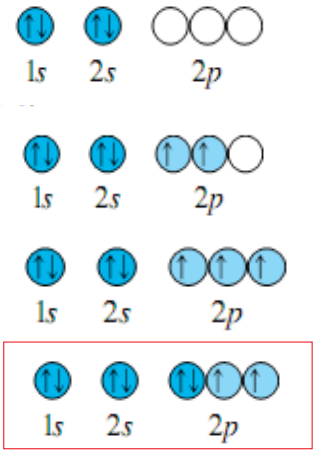


49

49

Hund's Rule

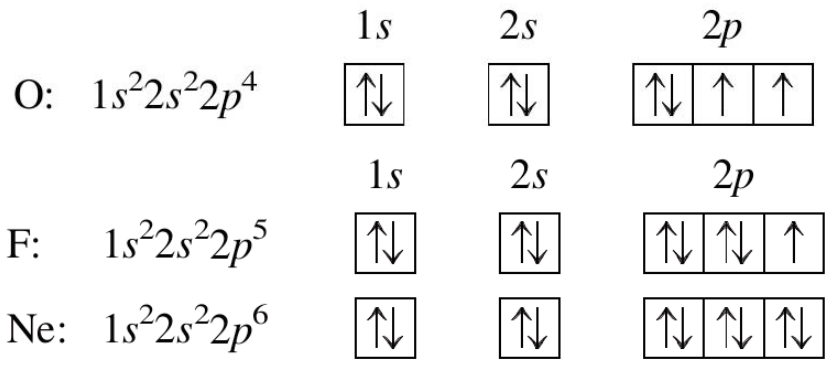
Example: *The Orbital diagram for the ground state of the Oxygen atom(Z=8)*



50

50

Hund's Rule



51

Class Guided Practice Problem

- Write the electron configurations for the following atoms: (a) Cs (z=55) and (b) Ni (z=28)

Class Practice Problem

- Write the electron configurations for the following atoms: (a) Se (z=34) and (b) Pb (z=82)

52

Problem

Use the building-up principle to obtain the configuration for the ground state of the Gallium atom (Z=31). Give the configuration in the complete form (do not abbreviate for the core). What is the valance-shell configuration?

Solution

➢ Using the aufbau principle and fill the subshells with electrons to distribute

$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^{10}$ $4p^1$

➢Or rearrange the subshells by shells:

$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $3d^{10}$ $4s^2$ $4p^1$

→ **Valance-shell: $4s^2$ $4p^1$**

53

53

Condensed Electron Configurations

- Neon completes the 2p subshell.
- Sodium marks the beginning of a new row.
- So, we write the condensed electron configuration for sodium as
  - Na: [Ne] $3s^1$
- [Ne] represents the electron configuration of neon.
- **Core electrons:** electrons in [Noble Gas].
- **Valence electrons:** electrons outside of [Noble Gas].

54

54

K: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ or $[\mathrm{Ar}]4s^1$

The inner electrons are known as core electrons

55

55

	Orbital Notation			Simplified Notation		
	1*s*	2*s*	2*p*			
$_3$Li	⇅	↑		$1s^2 2s^1$	or	[He] $2s^1$
$_4$Be	⇅	⇅		$1s^2 2s^2$		[He] $2s^2$
$_5$B	⇅	⇅	↑ _ _	$1s^2 2s^2 2p^1$		[He] $2s^2 2p^1$
$_6$C	⇅	⇅	↑ ↑ _	$1s^2 2s^2 2p^2$		[He] $2s^2 2p^2$
$_7$N	⇅	⇅	↑ ↑ ↑	$1s^2 2s^2 2p^3$		[He] $2s^2 2p^3$
$_8$O	⇅	⇅	⇅ ↑ ↑	$1s^2 2s^2 2p^4$		[He] $2s^2 2p^4$
$_9$F	⇅	⇅	⇅ ⇅ ↑	$1s^2 2s^2 2p^5$		[He] $2s^2 2p^5$
$_{10}$Ne	⇅	⇅	⇅ ⇅ ⇅	$1s^2 2s^2 2p^6$		[He] $2s^2 2p^6$

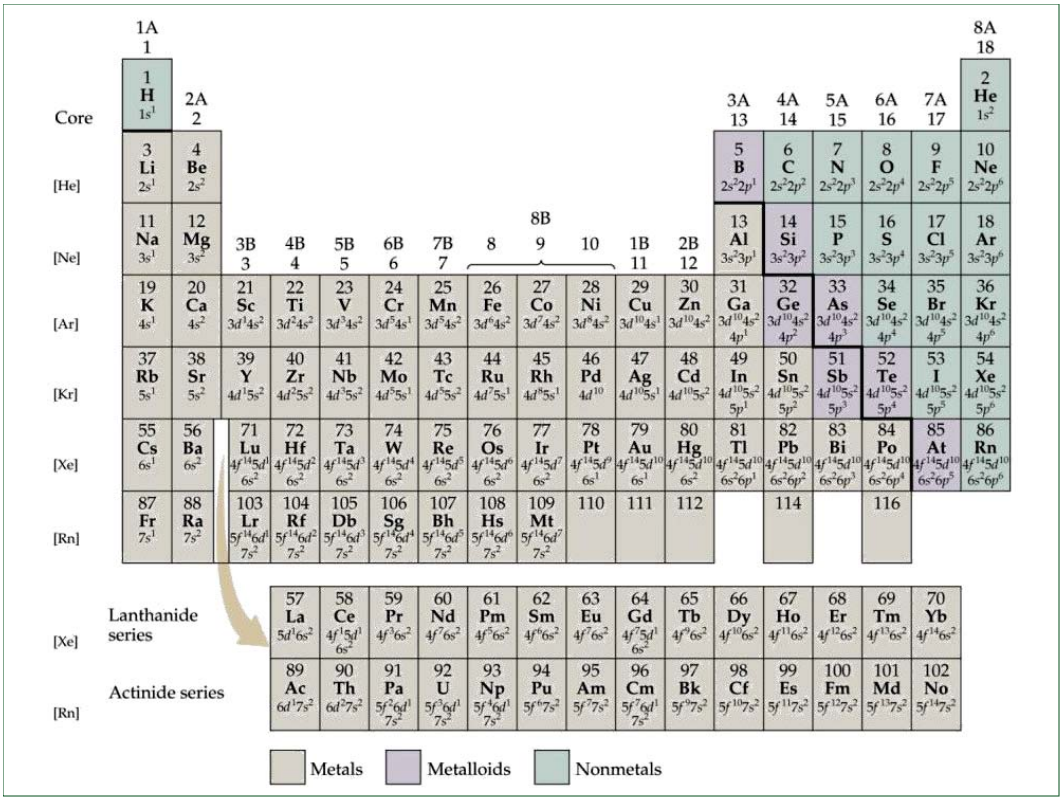
56

56

	Orbital Notation			
		3*s*	3*p*	Simplified Notation
$_{11}$Na	[Ne]	↑		[Ne] $3s^1$
$_{12}$Mg	[Ne]	↑↓		[Ne] $3s^2$
$_{13}$Al	[Ne]	↑↓	↑ _ _	[Ne] $3s^2 3p^1$
$_{14}$Si	[Ne]	↑↓	↑ ↑ _	[Ne] $3s^2 3p^2$
$_{15}$P	[Ne]	↑↓	↑ ↑ ↑	[Ne] $3s^2 3p^3$
$_{16}$S	[Ne]	↑↓	↑↓ ↑ ↑	[Ne] $3s^2 3p^4$
$_{17}$Cl	[Ne]	↑↓	↑↓ ↑↓ ↑	[Ne] $3s^2 3p^5$
$_{18}$Ar	[Ne]	↑↓	↑↓ ↑↓ ↑↓	[Ne] $3s^2 3p^6$

57

57



58

Electron Configurations and the Periodic Table

- The periodic table can be used as a guide for electron configurations.
- The period number is the value of n.
- Groups 1A and 2A (1 & 2) have the s-orbital filled.
- Groups 3A - 8A (13 - 18) have the p-orbital filled.
- Groups 3B - 2B (3 - 12) have the d-orbital filled.
- The lanthanides and actinides have the f-orbital filled.

59